AgentDescent: Gradient Descent Where the Parameters Are Agents

A Parallel, Asynchronous, Merge-Based Framework for Self-Evolving Agent Artifacts

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Abstract

Self-improving agent systems evolve their own skills, prompts, and harnesses, but largely through a *serial* loop that bounds improvement throughput at $O(1/T_{\text{iter}})$. **AgentDescent** transplants the distributed-training stack onto this problem: artifacts as parameters, diffs with attached evidence as gradients, a version-vectored ledger as the parameter server, and merge decisions as optimizer steps. Because diffs do not add, aggregation becomes a discrete-space optimization — conflict resolution, fusion, and statistical acceptance under a per-artifact Beta posterior. Eighteen published algorithms port onto the engine as plug-ins and run unchanged under serial, synchronous, and barrier-free scheduling.

Measured on live models: synchronous parallelism gives a median 1.36× end-to-end speedup across eleven algorithms (11/11, p=0.001), while barrier removal reaches 6.28× effective concurrency where the gate is expensive and nothing in aggregate elsewhere. For merging we derive a precondition in closed form — the engine proposes only from failed rollouts, so a near-saturated artifact starves the merge step however finely its keys are cut — and on the workload that condition prescribes, merging delivers *acceleration at no measurable quality cost*: indistinguishable from both baselines at six seeds, for 11–32% fewer model calls. It is not shown to improve quality. Code and reproduction commands are open source.

1 Introduction

A growing body of work shows that large-language-model agents can improve their own artifacts: prompts [4, 7], in-context playbooks [41], skill libraries [32, 43], agentic workflows [11, 40], and their own code [25, 36, 39]. Most of these systems share an execution structure inherited from the Gödel machine [27]: a serial improvement loop in which one candidate modification is proposed, evaluated, and accepted or rejected before the next is considered. If one iteration takes T_{iter} wall-clock seconds, improvement throughput is bounded at one accepted change per T_{iter}, independent of available compute. Notable exceptions parallelize at the *population* level — FunSearch and AlphaEvolve run asynchronous samplers over island archives [24, 26] — but their candidates remain independent: concurrent work is selected among, not merged.

Distributed deep learning encountered the same bottleneck and resolved it with a now-standard stack: data-parallel workers computing gradients concurrently, parameter servers aggregating them [19], staleness-tolerant asynchronous updates [23], per-parameter adaptive steps [17], and gradient surgery for conflicting updates [37]. This paper develops the thesis that the same stack, suitably reinterpreted, applies to agent self-evolution — and identifies precisely where the reinterpretation must depart from the original.

AgentDescent (Fig. 1) instantiates the mapping. A library of evolvable artifacts plays the role of the parameter tensor; a diff annotated with an *evidence card* (the rollout that motivated it) plays the role of

a gradient; a git-backed, version-vectored ledger plays the role of the parameter server; and the optimizer step is an *aggregator* that merges concurrent diffs. N workers roll out tasks against snapshots of the library and propose diffs in parallel; an asynchronous merger aggregates them, targeting $O(N/T_{\text{iter}})$ improvement throughput.

The mapping is productive precisely because of where it breaks. *Gradients add; diffs do not.* Two gradients can always be summed; two textual edits to the same artifact may be complementary, redundant, or contradictory. Aggregation is therefore not averaging but a decision procedure — staleness filtering, conflict resolution, fusion, and a statistical acceptance test (Section 4). A second disanalogy is equally consequential: training code is not modifiable by the gradients it computes, whereas a self-evolving system can in principle rewrite its own verifier. AgentDescent therefore enforces a layered governance model in which safety-critical components are frozen (Section 6).

Contributions.

1. **A system design and a discrete-space aggregator** mapping the distributed-training stack onto parallel self-evolution — per-diff bounded staleness, conflict projection, fusion, Beta-posterior acceptance, dual-branch promotion — with a formal account of where the analogy holds and where it must break (Sections 3 and 4).

2. **An efficiency analysis identifying three sources of speedup** (Section 5): rollout overlap, barrier removal, and — specific to merge-based evolution — reduced validation cost, in which merging collapses k acceptance tests into one and *reflective merge* extends the reduction to artifacts whose concurrent proposals conflict. The three do not compose unconditionally, and we characterize the regimes in which each pays.

3. **An extensibility mechanism** under which published algorithms are ported as plug-ins rather than forks (Sections 3.3 and 7): a diff-encoding `strategy` plus an optional `aggregator_factory`, or a declarative policy bundle. Eighteen ports run under all three schedulers without modification.

4. **A precondition for merging, in closed form** (Eq. (5)): because the engine proposes only from failed rollouts, the fraction of rounds with two diffs to fuse is $1-(1-f)^N-Nf(1-f)^{N-1}$ in the incumbent's failure rate f and the worker count N. It retrodicts two of our null results, prescribes the workload that avoids them, and bounds fusion width from below where the trust region and the round count bound it from above.

5. **A live-model evaluation** (Section 8) reported symmetrically: an eleven-algorithm scheduler study in which synchronous parallelism wins for all eleven methods (median 1.36× end-to-end) while barrier removal shows no aggregate further gain (0.99×), against a three-seed case study in which it reaches a median 6.28× on a gate-heavy workload — with the regime distinction analyzed; direct observation of the validation-cost reduction as falling model-seconds per rollout at a pinned budget; quality results under the most aggressive schedule; and an equal-budget merge-versus-selection comparison reported, with analysis, as not yet statistically separated.

2 Related Work

Self-evolving agents span prompt evolution [4, 7], context and playbook evolution [41], skill libraries [32, 43], workflow and harness search [11, 40], self-modifying code [25, 36, 39], evolutionary program search [18, 22, 24, 26, 28], and label-free self-play [2, 12, 42], with reflection [21, 29] as a shared inner proposal mechanism. Prompt optimization more broadly includes meta-prompting and search methods — APE [44], OPRO [34],

Table 1: The correspondence between distributed training and AgentDescent. The final row is a deliberate disanalogy, enforced rather than elided.

Model training	**AgentDescent**
parameter tensor θ	library of `Evolvable` artifacts
gradient g	`Diff` + `EvidenceCard`
parameter server	git-backed, version-vectored `Ledger`
optimizer step	`Aggregator` merge decision
per-parameter adaptive step (Adam)	per-artifact Beta-posterior acceptance
bounded staleness (async SGD/RL)	per-diff lag η + rebase re-verification
EMA / weight averaging	`dev`/`stable` dual branch
training code (not self-modifiable)	L0 frozen governance layer

DSPy [16], EvoPrompt [10] — and TextGrad [38], which also frames LLM feedback as a gradient analogue: TextGrad backpropagates textual critiques through a single computation graph, whereas AgentDescent's "gradients" are concurrent keyed diffs merged into one lineage under statistical acceptance. Population-based training [14] and the island architectures of FunSearch and AlphaEvolve parallelize by maintaining independent candidates and selecting among them; AgentDescent instead merges concurrent edits at the diff level. Most of the systems above publish a serial outer loop; AgentDescent treats that loop as one scheduling regime among three. On the systems side, AgentDescent borrows its vocabulary from distributed training: parameter servers [19], lock-free and bounded-staleness asynchrony [1, 8, 23], per-parameter adaptivity [17], gradient surgery [37], weight averaging [33], and tensor/pipeline parallelism [13, 30]. Concurrent work on parallel self-evolution (barrier-free evolution loops; co-evolutionary verification [3]) indicates that the throughput premise is an open hypothesis; our contribution is the specific synthesis — concurrent, staleness-bounded, conflict-resolved diff-level merge over a versioned ledger — together with the measurement harness required to evaluate it.

3 Problem Setup and System Design

3.1 Problem formulation

Let $\mathcal{A}$ denote the space of *artifact libraries*: finite keyed collections $A = \{a_k\}$ of versioned textual or executable artifacts (skills, prompts, playbooks, harness modules, file trees). An agent parameterized by A induces a policy π_A; given a task distribution $\mathcal{D}$ and a bounded reward $r \in [0, 1]$, the objective is

$$A^\star \;=\; \arg\max_{A \in \mathcal{A}} \; \mathbb{E}_{x \sim \mathcal{D}}\big[\, r(\pi_A(x),\, x)\big]\,, \tag{1}$$

optimized from rollout evidence only: $\mathcal{A}$ is discrete, and no gradient of r with respect to A exists. The serial baseline — propose one candidate A', evaluate on a held-out split D_{val}, accept if better — performs one accepted modification per iteration time T_{iter}. AgentDescent's premise is that with N concurrent proposers and merge-based aggregation, accepted-modification throughput can approach $O(N/T_{\text{iter}})$ without sacrificing the acceptance discipline.

A *diff* d is a keyed partial edit over A, produced from a rollout by a *strategy* S that maps a natural-language proposal to operations on keys (rule slots, playbook bullets, file paths). The key structure is what makes two diffs comparable: edits on disjoint keys *fuse*; edits on the same key *conflict*. Every diff carries an *evidence card* $e = (x, \tau, r, v_{\text{base}})$ recording the task, rollout, reward, and the library version the proposer observed.

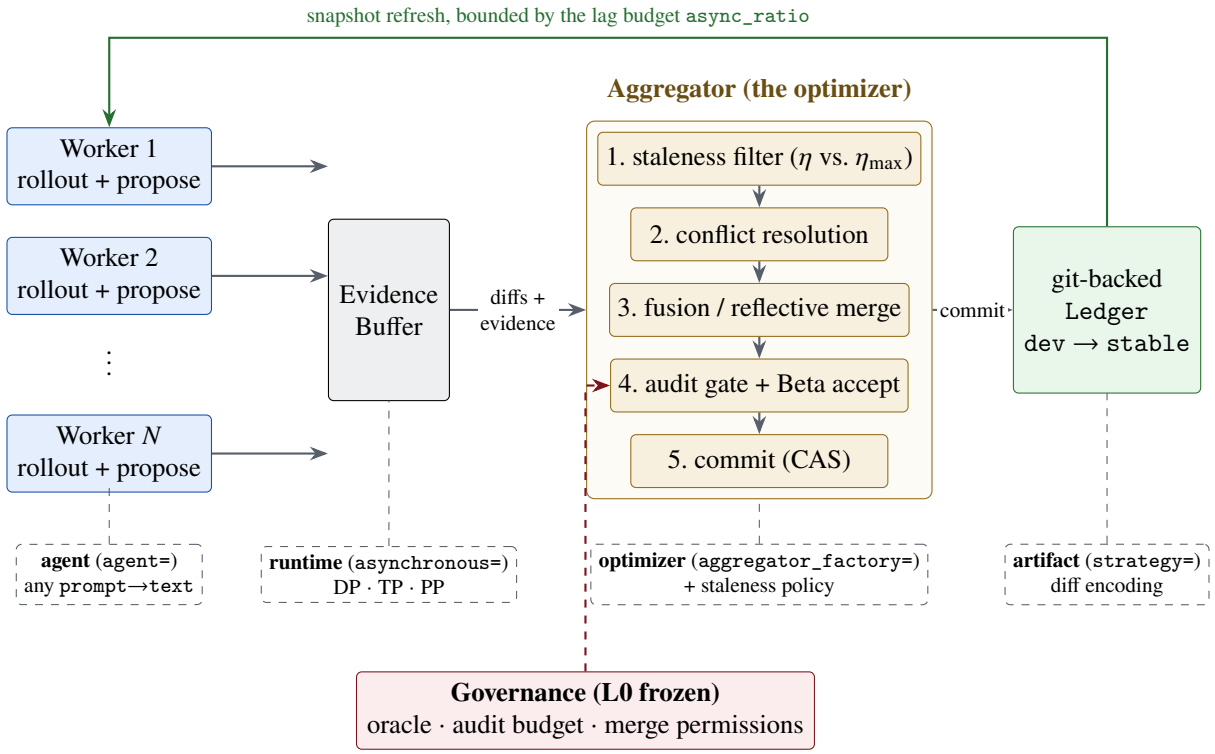

Figure 1: AgentDescent architecture. **Solid, top:** the fixed data path — N workers roll out tasks against ledger snapshots and emit diffs with evidence cards into a thread-safe buffer; a dedicated aggregator thread runs the five-stage merge pipeline and commits winners to the git-backed ledger; the green return edge is the only feedback path, bounded by the lag budget. **Dashed, bottom:** the pluggable seams of Section 3.3, each selected by one keyword argument of `evolve()`. **Red:** the frozen L0 governance layer, deliberately not a seam.

3.2 System components

Table 1 summarizes the correspondence and Fig. 1 the resulting architecture. The **ledger** is the parameter server: a git-backed store with per-artifact version vectors, compare-and-swap (CAS) commits, and a dual `dev`/`stable` branch pair; `stable` is promoted after K regression-free rounds on `dev`, an EMA-style confirmation in which a new commit restarts the counter. **Workers** hold snapshots of the library, roll out tasks, and emit (diff, evidence) pairs into a thread-safe buffer. The **aggregator** consumes the buffer and performs merge decisions (Section 4). Version vectors make the staleness of any diff computable at merge time:

$$\eta(d) \;=\; \max_{a \,\in\, \text{touched}(d)} \big(v_{\text{head}}(a) - v_{\text{base}}(a)\big). \tag{2}$$

3.3 Pluggable seams

Each component in Fig. 1 sits behind an explicit interface, selected by one keyword argument of the single entry point `evolve()`: the *agent layer* (any `prompt` → `text` callable is a completion — Claude, OpenAI-compatible endpoints, a tool-using CLI agent, or plain `run`/`reward`/`propose` functions), the *strategy* (diff encoding), the *aggregator* (`aggregator_factory`), the *staleness policy*, the *runtime* (synchronous or barrier-free), the *parallelism scheme* (data/tensor/pipeline [13, 30]), and the *executor and sandbox*. Selection and acceptance rules are likewise named policy classes. Section 7 uses these seams to port eighteen published algorithms without modifying the engine. The single deliberate exception is governance (Section 6): the L0 layer exposes no seam, since a self-modifying system must not be able to replace its own evaluator.

4 The Aggregator as a Discrete-Space Optimizer

Evidence cards are bucketed per artifact; when a bucket reaches batch size B (or a timeout, so cold artifacts are not starved), the aggregator executes one optimizer step, summarized in Algorithm 1: **(1) staleness filtering** by Eq. (2) under the active policy (Section 5.2); **(2) conflict resolution** — syntactic (overlapping hunks) and semantic (contradictory operations) conflicts are detected, and contradictions are projected out in the manner of PCGrad [37], retaining the better member of each pair on a shared held-out subset; **(3) fusion** — complementary survivors are merged into one union candidate (cf. model soups [33]), with *reflective merge* handling contradictions that a keyed union cannot (Section 5.3); **(4) auditing** — merges with high blast radius or low trust are routed through an oracle with veto power; **(5) statistical acceptance and commit**. Let (s_c, f_c) and (s_b, f_b) be the candidate's and the incumbent's successes and failures on the held-out split D_{val}. The candidate commits (by CAS on `dev`) iff

$$\Pr[\theta_c > \theta_b] \;>\; 1 - \delta_v, \qquad \theta_c \sim \text{Beta}(1 + s_c,\, 1 + f_c), \quad \theta_b \sim \text{Beta}(1 + s_b,\, 1 + f_b), \tag{3}$$

with independent uniform priors and the probability estimated by Monte Carlo, matching the released implementation — a per-artifact posterior test rather than a point threshold, the counterpart of per-parameter adaptivity in Adam [17]. The confidence requirement δ_v is annealed over the artifact's version v (the learning-rate-decay analogue) and a trust region caps diff size.

One risk of this discipline deserves explicit statement: the gate reuses a single fixed D_{val} across many acceptance decisions (hundreds of proposals per run in Section 8.3), so accepted diffs are selected extreme statistics on D_{val} and their measured gains are optimistically biased — the standard adaptive-data-analysis concern. Two mitigations are in place and one is not: all quality results in Section 8 are reported on a disjoint *test* split the gate never sees; the trust region and annealed δ_v limit how much any single decision can exploit D_{val}; but no reusable-holdout or D_{val}-rotation mechanism is implemented. We therefore measure the resulting bias directly (Section 8.4): across 33 runs the gain the gate observed on D_{val} exceeds the gain realized on test by 0.038 on average, a small but statistically detectable optimism that a reader should subtract from any D_{val}-reported figure.

A natural objection to merge-based evolution is that two individually beneficial edits may be jointly harmful. Two properties bound the risk: the acceptance test Eq. (3) scores the *fused* candidate on the full held-out split, so a harmful fusion cannot commit; and the union is a superset of every constituent diff, so committing it discards no proposal. An optional *fusion tournament* additionally ranks the union against its constituents; it is disabled by default (it costs one additional held-out sweep per candidate against a recoverable gain) but serves as the measurement instrument for the objection, reporting win rate, the losing tail, and a `contested` counter that records whether a fused candidate was in fact constructed and ranked — so a run in which fusion never occurred cannot be reported as a merge win. We deliberately publish no headline win rate: the quantity is a property of the artifact's key granularity and of proposal overlap, not of the mechanism (Section 8.5 reports it in situ for one workload).

5 Efficiency Analysis: Three Sources of Speedup

Wall-clock time in a self-evolution loop decomposes into rollout time, synchronization time, and held-out validation time. AgentDescent addresses each with a distinct mechanism; the third is specific to merge-based evolution.

5.1 Overlapping I/O-bound rollouts

Agent rollouts are dominated by time spent waiting on a model endpoint, so thread-level concurrency suffices to overlap them (Section 8.1 measures 5.8× at eight threads on real API calls). The synchronous data-parallel

runtime shards tasks across N workers over a common snapshot and merges at a round barrier. The barrier, however, imposes a ceiling independent of N: with i.i.d. rollout latencies t_i,

$$T_{\text{round}}^{\text{sync}} \;=\; \mathbb{E}\Big[\max_{i \le N} t_i\Big] \;+\; T_{\text{gate}}, \tag{4}$$

and reasoning-model latencies are heavy-tailed — a short answer and a long deliberation are the same call — which maximizes the gap between $\mathbb{E}[\max_i t_i]$ and $\mathbb{E}[t]$. The second term is a serial region in which all workers idle while the gate evaluates.

5.2 Barrier-free execution under bounded staleness

The asynchronous runtime (Algorithm 1) removes the round structure: workers produce continuously, a dedicated merger consumes continuously, and per-rollout cost approaches $\mathbb{E}[t]$ — the no-barrier discipline of asynchronous RL systems [1, 8] transferred to self-evolution. Its cost is staleness, which Eq. (2) makes first-class and per-diff, bounded by two mechanisms. A *lag budget* ρ (`async_ratio`) limits drift at the source: a worker refreshes its snapshot once v_{head} exceeds its snapshot version by more than ρ, and backpressure forces a global resynchronization if evidence accumulates while nothing commits. A *staleness policy* disposes of stale diffs at the merger: Full admits them unconditionally (sound only where diffs compose); Guarded rebases within $\eta \le \eta_{\max}$ and discards beyond, in the spirit of bounded-staleness RL [8], paying for safety in discarded rollouts; Reflective always rebases and *re-verifies* — a stale diff is treated as unproven rather than invalid, and is discarded only if its claimed improvement fails to hold on the new head. Because ρ is denominated in artifact versions, its appropriate value scales with rollout cost; the runtime emits a warning when a run discards the majority of its evidence.

5.3 Reducing validation cost

The third source of speedup is unavailable to the first two mechanisms: the gate itself. Every acceptance decision costs one sweep over D_{val}, and on live models this dominates the merger's critical path: profiled on a live GEPA/HotpotQA run (N=4, `GLM-5.2`), the merge thread's gate time *equals* its total busy time (560.1 s of each within a 740 s process, i.e. merger occupancy $\ge$ 76%, all of it evaluation). A serial or selection-based loop with m proposals pays $m \cdot |D_{\text{val}}|$ gate evaluations. Merging changes the count: a round's k surviving diffs fuse into one candidate, so per-round gate cost falls from $k \cdot |D_{\text{val}}|$ to $|D_{\text{val}}|$; a diff discarded by the staleness filter never reaches the gate and costs zero evaluations; and in the asynchronous runtime, cards arriving during a merge enlarge the next batch rather than triggering additional merges. The effect is visible directly in the RQ1 experiment (Fig. 3): model-seconds per rollout fall 37.9 $\to$ 27.7 $\to$ 23.8 across the serial, synchronous, and asynchronous arms at a pinned rollout budget. Two costs partially offset the saving and are stated for balance: the Reflective staleness policy's re-verification spends additional gate evaluations, and a model-written reflective merge spends one further model call per contested fusion. Residual gate cost parallelizes independently of the worker pool (`eval_concurrency`), saturating at $|D_{\text{val}}|$.

The reduction presupposes that proposals *can* fuse, which fails exactly when the artifact's key space is too coarse. An artifact held in a single key — e.g. a prompt as one instruction slot, GEPA's setting [4] — makes every pair of concurrent proposals a write–write conflict; a keyed union then degenerates to retaining the better single proposal, i.e. best-of-n selection at best-of-n validation cost. *Reflective merge* restores the reduction: a model is prompted to write the *union* of the contradicting proposals — one candidate preserving each proposal's intent — and the union passes through the same acceptance test as any candidate, so the model proposes and the gate decides. Key granularity is necessary but not sufficient, and the missing condition is quantifiable. The engine proposes only from a rollout that *failed* — a solved task yields no diff — so if the incumbent fails a fraction f of tasks and N workers run per round, the probability that a round produces at

Algorithm 1 Barrier-free evolution (`async_evolve`); the synchronous runtime is the special case with a round barrier between the two loops. S is the strategy of Section 3.1. Backpressure has two parts: `buffer.put` blocks while the un-merged intake exceeds the lag budget, and a stall detector (evidence arriving while nothing commits) forces a global resynchronization. In the current implementation a single merger thread performs all commits, so the CAS never contends; multi-merger operation, where it would, is future work.

Worker i **(in parallel,** $i = 1..N$**):**
loop
  if $v_{\text{head}} - v_i > \rho$ **then** refresh snapshot $A_i \leftarrow$ ledger head; $v_i \leftarrow v_{\text{head}}$
  end if
  sample task x; rollout $\tau \leftarrow \pi_{A_i}(x)$; reward $r \leftarrow r(\tau, x)$
  $(d, e) \leftarrow S.\text{propose}(A_i, x, \tau, r)$; buffer.put$(d, e)$ ▷ blocks on backpressure
end loop
Merger (single thread):
loop
  $\mathcal{B} \leftarrow$ buffer.drain(); bucket $\mathcal{B}$ by artifact
  for each bucket with $|\mathcal{B}_a| \geq B$ or timed out **do**
    $\mathcal{B}_a \leftarrow \{\, d : \text{policy}(\eta(d), \eta_{\max}) \neq \textsc{discard} \,\}$, rebasing where directed ▷ Eq. (2)
    resolve conflicts (keep better of each contradicting pair); $u \leftarrow$ fuse(survivors)
    if survivors contradict on a key **and** reflective merge enabled **then** $u \leftarrow$ model-written union
    end if
    if audit passes **and** Eq. (3) holds on D_{val} **then** CAS-commit u to `dev`
    end if
  end for
  promote `dev` $\rightarrow$ `stable` after K regression-free rounds
end loop

least two diffs, and therefore anything to fuse, is

$$P_{\text{fuse}}(f, N) \;=\; 1 - (1-f)^N - Nf(1-f)^{N-1}. \tag{5}$$

This is the design condition merging must satisfy before it can be compared with selection at all, and it is easy to violate: a near-saturated artifact starves the merge step no matter how finely its keys are cut. Section 8.5 measures both sides of it. It is the spatial counterpart of the Reflective staleness policy (carry intent onto a changed base; let measurement decide), and the `contested` counter reports whether fusion genuinely occurred, so the saving is never claimed where only selection took place.

6 Governance by Blast Radius

Artifacts are assigned to layers by a declared `blast_radius`: **L2** (skills, prompts) merges under the full asynchronous pipeline; **L1** (harnesses, verifiers) serializes in-flight changes and routes every merge through the oracle; **L0** (the oracle, audit budget, merge permissions, safety constraints) is frozen and read-only to the loop. The frozen layer enforces the second disanalogy of Table 1: without it, a self-referential loop can converge to a verifier that accepts its own output. The same layering protects executed agent code, which runs behind a frozen test suite the candidate cannot rewrite; pristine tests are overlaid after workspace materialization, so weakening them at run time is likewise ineffective.

7 Case Study: Porting Eighteen Self-Evolution Algorithms

Because every component sits behind a seam (Section 3.3), porting a published algorithm requires a small set of plug-ins rather than a fork. Two routes cover all eighteen ports.

Route 1: a strategy, plus an optimizer where needed. A *benchmark-faithful* port supplies a `strategy` encoding the paper's artifact as keyed diffs and, only when its search differs from the default pipeline, an `aggregator_factory`. The seven faithful ports illustrate the scale of the required code: ACE [41] is a playbook strategy whose curator *is* the default aggregator; GEPA [4] substitutes a per-instance Pareto optimizer; ADAS [11] and DGM [39] plug in keep-all archives with their own parent selection; EvoSkill a bounded top-K frontier; SkillOpt a strict edit gate;[1] OpenEvolve [28] MAP-Elites islands [22]. Parent-selection and acceptance rules are extracted as named policy classes, so subsequent ports reuse them.

Route 2: a declarative policy bundle. The eleven microports and analogues (PromptBreeder, AFlow, Self-Refine, Reflexion, SICA, Gödel Agent, Voyager, SkillWeaver, Absolute Zero, R-Zero, Agent0) define no new classes: each is a `MethodPolicy` — selection rule, memory, acceptance shape, genome — over one shared runner, inheriting a common budget contract, dataset layer, and command line. A shared `--serial` flag reproduces the upstream algorithm's own semantics (one worker, serial gate) as the control that any parallelization claim requires, and a shared rollout budget pins compared arms to equal work.

Fidelity is declared per port — *benchmark-faithful*, *mechanism microport*, *self-edit analogue*, *environment analogue*, or *inference analogue* — follows the released code where it diverges from the paper, and documents infrastructure boundaries explicitly; analogues are not to be cited as benchmark reproductions. All eighteen ports run under serial, synchronous, and barrier-free scheduling without modification, which is what permits the controlled scheduler comparison of Section 8.2.

8 Evaluation

Setup. All results are obtained from live models on real datasets: `deepseek-v4-flash` and `GLM-5.2` (reasoning models) via an OpenAI-compatible endpoint, on each algorithm's own benchmark. Every number is produced by a named script in the repository with its exact reproduction command; absolute times depend on host and endpoint, so ratios are the reproducible quantity. We address four questions: **RQ1** — how much effective concurrency do the three mechanisms of Section 5 deliver against a faithful serial control? **RQ2** — does the speedup generalize across algorithms? **RQ3** — is learning behavior preserved under the most aggressive schedule, and how optimistic is the gate that decides it? **RQ4** — does merging accelerate without a quality penalty relative to best-of-N selection at equal budget?

8.1 RQ1: Effective concurrency against a faithful serial control

We first verify the premise of Section 5.1: eight threads over one pool reduce a real `GLM-5.2` API workload from 48.6 s to 8.3 s (5.8× in this run; 5.8–6.5× across three repeats, sub-linear under heavy-tailed latency), while the same threads on pure-Python CPU work yield 1.1× — rollout overlap is real, and the GIL does not limit I/O-bound rollouts.

The main experiment ports GEPA to its own benchmark (HotpotQA [4, 35]) with 16 rollouts pinned on every arm, three seeds, and a control that reproduces the upstream loop exactly: one worker *and* a serial gate. That control lands at 0.99, 1.00, and 1.00× overlap on the three seeds (Fig. 2), which is worth stating on its

[1] EvoSkill and SkillOpt are ports of released systems whose upstream papers and per-port departures are identified in the repository's port-fidelity documentation; we do not cite them here to avoid attributing a reconstruction to the wrong reference.

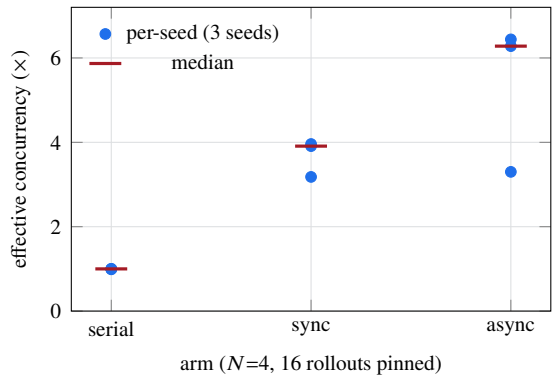

Figure 2: Effective concurrency (model-seconds ÷ wall-clock), GEPA on HotpotQA, `GLM-5.2`, 16 rollouts pinned per arm, three seeds. The serial control reproduces the published loop (one worker, serial gate) and lands at 0.99–1.00× on every seed — the control property is a stable one, not a single-run coincidence. Synchronous parallelism spans 3.18–3.96×; barrier removal spans 3.30–6.44× but is by far the widest arm, and it reaches those figures while overrunning the pinned budget (19 rollouts against 16).

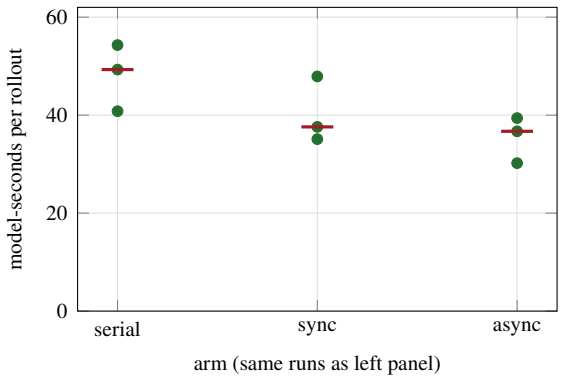

Figure 3: Validation-cost reduction in the same runs (Section 5.3): model-seconds spent per rollout at a pinned rollout budget, per seed, with medians marked. The parallel arms evaluate less per unit of work — median 49.3 → 37.6 → 36.7 — because reflective merge collapses each round's admissions into one gate sweep and stale proposals are discarded before they are scored.

own: the property that makes it a faithful control is stable, not a fortunate single run. Against it, synchronous parallelism spans 3.18–3.96× effective concurrency (median 3.91) and barrier removal spans 3.30–6.44× (median 6.28); median wall-clock falls 790 → 152 → 116 s. Figure 3 shows the validation-cost mechanism of Section 5.3 in the same runs, and it replicates across seeds: model-seconds per rollout fall from a median 49.3 (serial) to 37.6 (synchronous) to 36.7 (asynchronous) at the pinned budget — the parallel arms not only finish sooner, they evaluate less per unit of work. Independent endpoint probes sustain 7× (matched width and request length) to 10× (wider, shorter requests) concurrency, so no arm is provider-limited.

Three qualifications travel with these numbers, and they matter more than the headline. First, *the asynchronous arm is not paying the same bill*: lacking a round boundary it ran 19 rollouts against the pinned 16 on every seed, so part of its concurrency is work the other arms did not perform. Second, *it is also the least stable arm* — a 3.30–6.44× spread against synchronous parallelism's 3.18–3.96×. Third, and most important, *none of this is a quality claim*: median held-out test score is 0.55 (serial), 0.60 (synchronous), 0.50 (asynchronous) across the three seeds, on a 20-item split whose resolution is 0.05, so the arms are indistinguishable in quality and the fastest arm is not the best one. One synchronous cell (seed 0) also ended early on an endpoint error at 12 of 16 rollouts and is included as measured.

These figures supersede a single-run measurement of 1.85 × /3.22× on a different model reported in an earlier draft. Two differences account for the gap and neither is a correction of the other: this run uses `GLM-5.2` rather than `deepseek-v4-flash`, and its parallel arms run the gate at `eval_concurrency=16`. The second is the more instructive: with the gate itself parallelized, the round barrier costs much less, which both raises the synchronous arm and shrinks what barrier removal has left to recover — exactly the regime dependence that Section 8.2 finds across eleven algorithms. An earlier draft additionally reported a width-8 variant with a 28% model-call saving; it is withdrawn because its serial arm ran with a concurrent gate (overlap 1.31×), violating the control definition above.

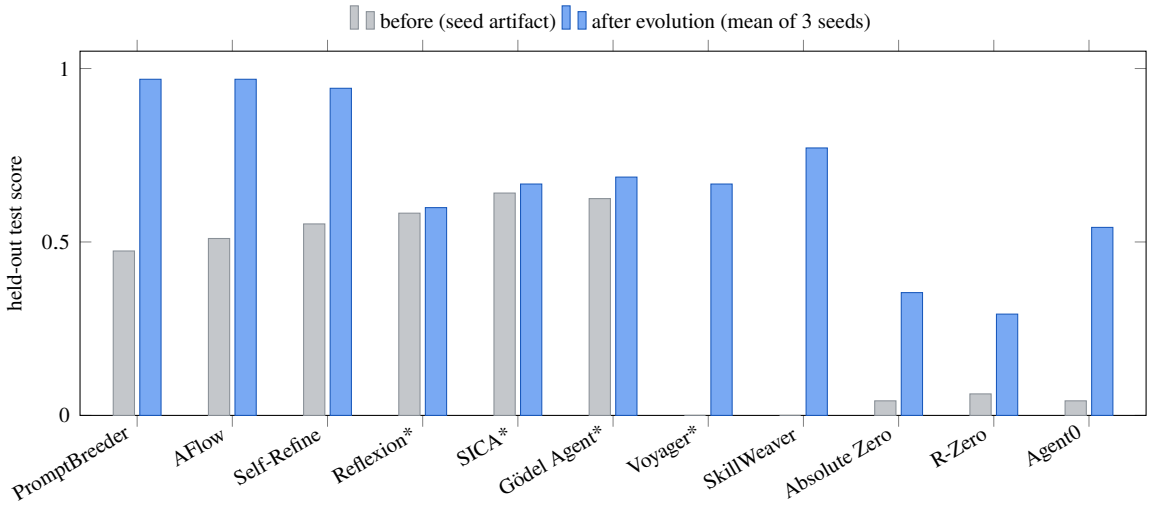

Figure 4: Held-out test score before and after evolution for the eleven microports and analogues, each run against `deepseek-v4-flash` under the *most aggressive* configuration — barrier-free async pipeline, 8 workers, Full staleness, 80 rollouts per seed, 3 seeds (465–2,069 model calls per seed). "After" bars are three-seed means; asterisked methods improved on two of three seeds (the rest on all three), and Reflexion's gain is within its domain's ±0.02 noise floor. Between 3 and 10 of the 240 proposals per method (80 × 3 seeds) commit — the gate admits only strict held-out improvements, so sparse acceptance is the shape of the mechanism, not a low yield. Bars are comparable only within a domain; per-seed spreads are tabulated in the repository documentation.

8.2 RQ2: Generality across algorithms

To test whether the RQ1 result generalizes beyond GEPA, the same scheduler comparison — serial, synchronous, barrier-free — is applied to all eleven declarative ports on `GLM-5.2` at an equal candidate budget and width 2, varying only the scheduler (three paired seeds per method; 99 observations). The results cut both ways, and we report both. *Synchronous parallelism generalizes*: paired over n=33 runs it outperforms the serial arm by a median 1.36× end-to-end and 1.89× within the engine window, with a 1.47× median time-to-quality (n=18 seeds reaching the quality bar). The win is unanimous across methods — 11 of 11 on both the end-to-end and engine-window medians, a two-sided sign test giving $p = 0.001$ in each case. *Barrier removal does not*: the asynchronous arm aggregates to 0.99× end-to-end and 0.97× engine-window relative to synchronous, and the per-method record is indistinguishable from chance — 5 methods faster, 5 slower, 1 tied end-to-end ($p = 1.00$), and 2 faster against 8 slower in the engine window ($p = 0.11$, i.e. if anything a slight cost). Its one positive signal is time-to-quality, a modest 1.10× median (n=15) concentrated in a few methods (SkillWeaver 1.25×, Self-Refine 1.24×, Reflexion 1.12×). The reconciliation with RQ1's 3.22× is the regime: barrier removal pays in proportion to how much heavy-tailed rollout latency and gate cost there is to hide, which GEPA's long rollouts and expensive Pareto gate maximize and these compact ports mostly lack; asynchrony helps time-to-quality only when a useful commit can land before a synchronous barrier would. Two caveats: this matrix was measured on the implementation preceding the ports' declarative restructuring (mechanisms and budgets unchanged by design; a rerun on current code is pending), and its per-run JSON is retained as a summarized report rather than raw. We report it despite the caveats because it is the only multi-algorithm evidence in either direction, and it bounds the asynchronous claim honestly.

8.3 RQ3: Quality preservation under the aggressive schedule

Throughput is meaningful only if learning is preserved, so quality is measured under the most aggressive configuration (barrier-free, 8 workers, FULL staleness). Figure 4 reports the eleven declarative ports at three seeds each. Read precisely: ten of eleven improve their three-seed mean test score beyond their domain's noise floor; Reflexion's +0.016 on GSM-Hard [5, 9] sits within that domain's ±0.02 floor and should be read as no change; and four methods (Reflexion, SICA, Gödel Agent, Voyager) improve on two of three seeds rather than all three. Gains are bounded by each benchmark's headroom — the three GSM8K ports converge to 0.94–0.97 and differ in cost rather than quality. What this establishes is that the aggressive schedule does not stop the ports from learning; a paired serial-schedule quality arm at equal rollout budget, which would establish that asynchrony costs no quality *relative to serial*, has not been run and we do not claim it. Among the benchmark-faithful ports, three representative single-seed results establish end-to-end operation (not effect sizes): ACE improves FiNER-139 [20] validation 0.719 → 0.766 over 120 rollouts while curating a ten-bullet playbook; DGM's best archived child repairs held-out vendored bugs at 0.906 versus its seed agent's 0.844 under a frozen `pytest` oracle (a compact SWE-bench-style setting [15]); SkillOpt improves a hard SearchQA [6] subset 0.053 → 0.211 with 3 of 43 proposed edits accepted. One port produced no quality number at all: ADAS's benchmarks were either saturated (MGSM) or cost-prohibitive per call (GPQA), and we state this rather than average over it. The highest-level interface (`evolve_skill`) on 40 HotpotQA rows raises held-out exact match from 0.167 to 0.583 over 338 model calls, committing one proposal.

8.4 RQ3, continued: how optimistic is the gate? The val–test gap

The acceptance test reuses one fixed D_{val} across every decision, so an accepted diff is a selected extreme statistic on that split and its measured gain should be optimistically biased (Section 4). The size of that bias is measurable from the runs already reported: each of the 33 runs behind Fig. 4 records both its D_{val} trajectory and its disjoint test score, so the per-run difference (test gain − val gain) estimates the optimism directly. Mean val gain is +0.395 and mean test gain +0.357, a mean gap of $\mathbf{-0.038}$ (median −0.031, s.d. 0.075); test gain falls short of val gain in 20 of 33 runs and exceeds it in 7, with 6 exact ties — a two-sided sign test on the 27 non-tied runs gives $p = 0.019$. The bias is thus real and directionally as predicted, but small relative to the gains themselves (roughly a tenth of the mean gain), and it does not threaten the sign of any result in Fig. 4, whose scores are reported on test throughout. The runs where it is largest are those with the coarsest reward signal (SkillWeaver, −0.125 to −0.250 on an eight-task test split), which is the expected place for a selected statistic to overfit.

8.5 RQ4: Merge-of-N versus best-of-N selection at equal budget

Throughput cannot distinguish merging from sampling-and-selecting, since N independent forks occupy N workers equally well; the discriminating quantity is held-out quality at an equal rollout budget. The largest such run to date — HotpotQA on `GLM-5.2`, three arms × three seeds at nine rollouts per seed, budget pinned in rollouts, reflective merge enabled so that contradicting proposals survive to fusion; 1,298 model calls (29 failing at the endpoint) and 11.3 hours of model time — yields a **null result**: merge-of-3 attains the highest median test score together with the widest spread, the intervals overlap, and the framework's own separation test declines to declare a winner. The diagnosis is more informative than the verdict: at this budget the fused union was constructed twice in the entire experiment — most merges saw a single surviving diff — so the experiment does not measure a merging deficit; it measures a regime in which merging rarely fired. It also confirms the key-granularity condition of Section 5.3 from the opposite direction: without reflective merge, the merge arm on this single-key artifact would have reduced to best-of-N selection. We therefore tested that diagnosis on a *multi-key* artifact: ACE's playbook on FiNER-139, keyed per bullet, so two workers editing different bullets compose by construction. Fusion still did not fire — over a 24-rollout, four-worker run the

aggregator held **2 tournaments and contested none**, one with a single candidate and none with contradicting candidates. The cause was not key granularity but Eq. (5): that artifact already scored 0.938 on the gate's split, so $f \approx 0.06$, and at N=4 the predicted fraction of rounds with anything to fuse is 2%. Two uncontested tournaments in six rounds is what a 2% rate looks like.

Equation (5) also prescribes the fix, so we built the workload it calls for. BBH [31] supplies tasks a strong model still fails, and `KeyedRules` supplies an artifact keyed by the category a failure diagnosis falls into (output format, method, verification, common errors). Crossing them needed no engine change — only a new `Workload`, because dataset and strategy are independent seams (Section 3.3). On `GLM-5.2` the seed artifact scores 0.100 on test, so $f \approx 0.9$, and at N=8 Eq. (5) predicts that essentially every round should have something to fuse.

It does. Over a 32-rollout run the aggregator held **3 contested tournaments** — the first in this project in which fusion actually ran — while the artifact improved from 0.100 to 0.400 on test, so the workload supplies both the failure rate and the headroom the question needs. The fused candidate **won 1 of the 3**, at a mean gain of −0.115 with two losses (worst −0.250); two of the three fell below the artifact they started from. On this workload, fusing a round's diffs was usually *worse* than keeping the better single one — and the acceptance gate refused exactly those, which is the safeguard of Section 4 behaving as designed rather than an argument against it.

With fusion firing on demand, the comparison the question has been waiting for becomes possible. We ran it at three seeds, and then — because the three-seed reading was favourable to merging — at three more. The second half did not replicate the first, and the combined result is the one we report: six seeds × three arms on `bbh-keyed`, 32 rollouts pinned on every arm, N=8, `GLM-5.2` (16,124 model calls, 28.3 hours of model time).

Table 2: Merge-of-N against best-of-N selection and a serial control at an equal rollout budget, on a workload built to satisfy Eq. (5), in two configurations. *Test* is a split no gate in any arm ever saw; *s.d.* is across seeds. *Oracle* is the best fork *on test*, an upper bound that requires the answer to select, against which the fork row reports what fork-and-select can actually ship. Wall-clock in the lower block is not comparable across arms (all nine runs shared one endpoint) and is omitted; calls are unaffected.

	arm	test min/med/max	mean	s.d.	oracle	calls	wall (s)
N=8, 32	serial	0.133 / **0.267** / 0.333	**0.250**	0.076	—	772	1483
	fork-of-8	0.067 / 0.200 / 0.300	0.189	0.085	0.267	1004	1766
	merge-of-8	0.033 / 0.217 / 0.400	0.194	0.130	—	**685**	**1157**
N=4, 96	serial	0.100 / 0.250 / 0.433	0.256	0.118	—	2070	—
	fork-of-4	0.133 / **0.300** / 0.400	**0.283**	0.105	0.333	2363	—
	merge-of-4	0.200 / 0.267 / 0.400	0.267	**0.067**	—	**1832**	—

Both blocks: six seeds. Upper: 4 rounds per run. Lower: 24 rounds per run.

The claim merging has to meet is non-inferiority, not superiority. The framework's premise is throughput — accepted improvement per unit wall-clock — so merging succeeds if it reaches the same quality faster and cheaper, and fails only if quality degrades. Read that way, the six seeds say: quality is indistinguishable (merge better on 2 seeds, worse on 3, tied on 1 against fork, mean paired difference +0.006; p=1.00), while merge is the cheapest arm in both currencies — 11% fewer model calls and 1.28× less wall-clock than serial, 32% fewer calls and 1.53× less wall-clock than fork.

A caution about the quality column, in both directions. On seeds 0–2 alone the merge arm posted a median of 0.267 against fork's 0.100, which would have read as merging's first favourable equal-budget result; seeds 3–5 reversed it (0.200 against 0.267). We report the pooled six because we would otherwise have published the favourable half. The same caution bounds the non-inferiority claim in the other direction: six seeds on a 30-task split detect only large effects, so "no measurable quality penalty" is what the data

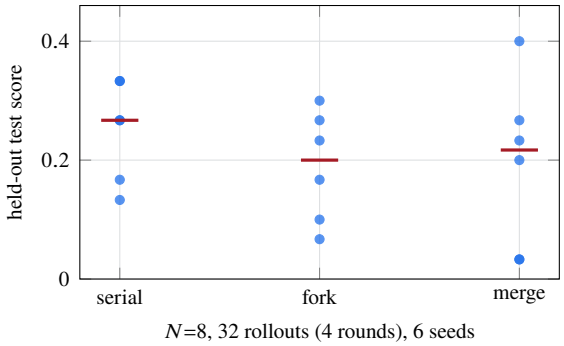

Figure 5: Fusion width 8. Per-seed test scores with medians marked. The three arms overlap; the merge arm has the *widest* spread (s.d. 0.130 against 0.076 and 0.085).

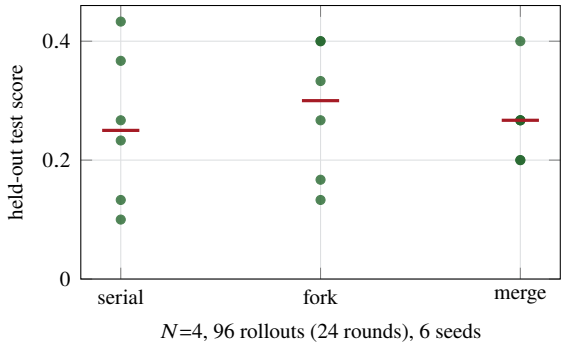

Figure 6: Fusion width 4, matched to the trust region, at a 3× budget, six seeds. The arms still overlap, but the merge arm is the *tightest* (s.d. 0.067 against 0.118 and 0.105) — the ordering that width 8 reverses.

supports, not "no quality penalty".

The width of a fusion is a design parameter, and eight was the wrong value. Equation (5) is satisfied almost equally by N=4 (99.6% of rounds have something to fuse) and N=8 (100%), but the two differ in what they do to the rest of the loop: at a fixed rollout budget N=8 halves the number of rounds, and therefore halves the opportunities to accumulate, while making each fused union an eight-way jump that the trust region of Section 4 is meant to prevent. Re-running at N=4 and a 3× budget (24 rounds), also at six seeds, changes the mechanism's behaviour in the predicted direction: the mean per-seed fusion gain rises from −0.135 to −0.045, and the merge arm's seed-to-seed standard deviation falls from 0.130 — the *worst* of the three arms — to 0.067, the *lowest*, against serial's 0.118 and fork's 0.105. Quality remains indistinguishable there (merge vs. fork 2–2–2, mean −0.017; vs. serial 2–3–1, mean +0.011; p=1.00 for both) and merge remains the cheapest arm (1832 calls against 2070 and 2363).

That variance reduction is the most interesting signal in these experiments, and it is the one that survived the check the quality claim failed. At three seeds it read as a factor of three (0.031); extending to six halved the effect to a factor of 1.6–1.8 (0.067 against 0.105 and 0.118), which is what regression to the mean looks like and is why we ran the extra seeds. What remains is an ordering that holds at both sample sizes and reverses when the fusion width is wrong, with a mechanism attached: merging averages several workers' evidence into one artifact, where selection keeps whichever single lineage the gate happened to prefer. We report it as a consistent secondary effect rather than a headline, because six seeds do not establish a variance ratio and we have not tested it on a second workload.

The fusion counters, across both configurations. At N=8 the aggregator held 10 contested tournaments over six seeds and the fused candidate won 2 (20%); at N=4, over six seeds, it held 21 and won 4 (19%), at a mean per-seed gain of −0.045 against width 8's −0.135. Fusion loses more often than it wins in both, and the acceptance gate refuses exactly those losses, which is the safeguard of Section 4 working as designed. The negative control holds throughout: across eighteen control runs the serial and fork arms held 823 tournaments and contested *none*, as their construction requires.

Our summary is therefore: **merging is an acceleration with no measurable quality penalty**, which is the claim the framework makes and the one these experiments support. It is not shown to improve quality, its fusions lose more contests than they win, and and its most promising property — lower seed-to-seed variance at a fusion width matched to the trust region — is a consistent secondary effect, not a headline: extending three seeds to six halved it.

9 Limitations

AgentDescent is a research reference implementation, and its evaluation has stated boundaries. (i) *The central claim is supported only in its weak form*: merging accelerates at no *measurable* quality cost, which at six seeds on a 30-task split means only that a large quality penalty was ruled out. Merging is not shown to improve quality, a favourable n=3 reading did not replicate (and is recorded rather than dropped), and the variance reduction we observe at N=4 halved between three seeds and six, so its size is not established even where its direction is. Fusion width is a parameter we got wrong before getting right, which is itself a warning: Eq. (5) constrains it from below, the trust region and the round count constrain it from above, and we have measured only two values of it. (ii) *Statistical strength*: the RQ1 ladder and the eleven-port quality results carry three seeds and are reported as spreads; the benchmark-faithful ports carry one seed each; the thread-overlap figure spans 5.8–6.5× over three repeats. Quality splits are small enough (20 items in RQ1) that no arm's quality is resolvable, which is why speed and quality are reported separately and no arm is claimed to be free. These establish that the mechanisms operate and in which direction, not effect sizes. (iii) *Scale*: the system is a single Python process — one merger thread (measured at ≥ 76% occupancy at N=4, all of it gate evaluation), thread workers, a local git ledger. Multi-machine operation does not exist, the CAS commit path never contends under a single merger, no worker sweep beyond N=8 has been run, and there is no failure-recovery story beyond endpoint retries; a merger-saturation model is future work. (iv) *Holdout reuse*: the gate reuses one fixed D_{val} across all acceptance decisions (Section 4); test splits are disjoint, but no reusable-holdout mechanism exists and the val–test gap of accepted diffs is not yet reported. (v) *Pending reruns*: the RQ2 matrix predates a port restructuring and awaits a rerun; turn-level checkpoint-resume of stragglers and a stratified tail-canary split are designed but unimplemented; difficulty parameters calibrated on one model did not transfer under a model swap, whereas benchmark-intrinsic difficulty did. (vi) *Correction trail*: several earlier measurements proved irreproducible or favorably defined and were re-measured and corrected in place — including, in an earlier draft of this paper, a width-8 experiment withdrawn for a flawed control and a misattributed per-rollout statistic (Section 8.1). Every number reported here is generated by a named script. We regard this trail as part of the contribution: the evaluation of a self-improving system should satisfy the standard its own acceptance gate imposes on diffs.

10 Conclusion

The distributed-training stack admits a coherent counterpart in agent self-evolution once its necessary disanalogy — diffs do not add — is taken seriously and merging is constructed as a discrete-space optimizer. Efficiency decomposes into three sources with different evidential status: rollout overlap and synchronous parallelism generalize across algorithms (median 1.36× end-to-end over eleven ports, all eleven improving); barrier removal is a regime-dependent win, reaching a median 6.28× effective concurrency on a gate-heavy workload while showing no aggregate gain across eleven compact ones; and reduced validation cost — merging collapses k acceptance tests into one — is visible directly as falling model-seconds per rollout at a pinned budget. Eighteen published algorithms adapt as plug-ins rather than forks. Whether merging also improves quality over best-of-N selection at equal budget remains open, but it is no longer untestable. The mechanism has a precondition we can state in closed form and satisfy on demand — enough failures per round that two diffs coexist — and on the first workload built to satisfy it the comparison finally runs. What it supports is the framework's actual claim rather than a stronger one: merging is an *acceleration at no measurable quality cost* — indistinguishable from both a serial lineage and best-of-N selection at six seeds, for 11–32% fewer calls and 1.28–1.53× less wall-clock. Its fused candidates lose more contests than they win, and the acceptance gate is what makes carrying a usually-worse union free. One further observation is more promising than anything else here and less certain: at a fusion width matched to the trust region, merging's seed-to-seed

variance is the lowest of the three arms at both three and six seeds, though the effect halved as seeds were added. Whether its real product is predictability rather than peak quality is the next experiment, not a claim of this paper.

Acknowledgments

The algorithm ports were contributed by `chendanyang` (the seven benchmark-faithful ports) and `cyanneko` (OpenEvolve and the eleven microports and analogues).

Reproducibility

All code is available at https://github.com/Birfy/agentdescent (MIT license; `pip install agentdescent`; the core engine has no required dependencies). Generating scripts, per result: thread overlap — `examples/efficiency.py --only gil`; RQ1 and the live merger profile of Section 5.3 — `bench/matrix_run.py --rows gepa`; RQ2 — `bench/candidate_methods.py`, measured on the pre-restructuring implementation (summarized report retained, raw per-run JSON not; rerun pending); RQ3 — each port's own command line over `examples/_method_runner.py`, the entry point that accepts the `--staleness full` configuration used, with the resolved configuration printed as one JSON line per run; the `evolve_skill` case — `examples/skill_evolution.py`; RQ4 — `bench/baselines_run.py --dataset bbh-keyed --fusion-tournament`, whose workload (BBH crossed with `KeyedRules`), width and budget sweeps, and fusion counters were added for this paper and ship with it. Every port supports `--dry-run` (no API key or model call) and carries an offline test suite; datasets are retrieved from canonical sources by a dependency-free data layer.

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